



Shop Learn Forums Bluesky Instagram Discord Github GBP Register / Login

Top notch support
57,844 reviews!

Next day delivery
order by 2pm

No fuss returns
contact support

Secure shopping
3000+ products

PIMORONI

Search...



£0
0 items

[raspberrypi](#) [Pico](#) [Pimoroni](#) [Adafruit](#) [Breakouts](#) [Badgeware](#) [Smart Home](#) [Kits](#) [Tools](#) [New](#)

Displays Cameras LED Strips Sensors Audio Feather Teensy Robotics Retro/Arcade Components

Type here to search...



PIMORONI

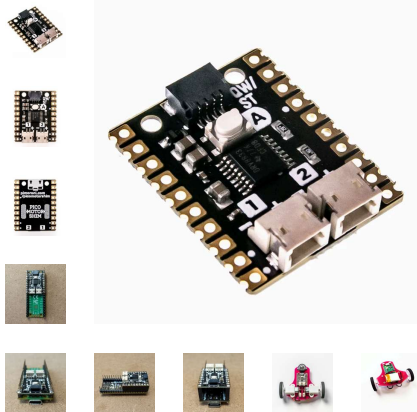
LTD Purveyors of Maker Goods EST. 2012

£0
0 items

[raspberrypi](#) [Pico](#) [Pimoroni](#) [Adafruit](#) [Breakouts](#) [Badgeware](#) [Smart Home](#) [Kits](#) [Tools](#) [New](#)

Displays Cameras LED Strips Sensors Audio Feather Teensy Robotics Retro/Arcade Components

Motor SHIM for Pico by Pimoroni



★★★★★ 7 reviews

login to add to wishlist

This neat little add-on board lets you drive two micro metal gearmotors from a Raspberry Pi Pico - tiny robots ahoy!



Ships today – if you order within 8 hours 36 minutes!

Motor SHIM for Pico

PIM617

£9.60

Add some pre-soldered motors



Micro Metal Gearmotor with Motor
Connector Shim (MCS) – 50:1 Sideways

+ £7.80

Motor connector cables



Motor Connector Shim Cable (pack of 2) –
50mm

+ £0.90

Brackets and adaptors



Long Micro Metal Gearmotor Bracket (pack
of 2)

+ £3.90

**This neat little
add-on board lets
you drive two
micro metal
gearmotors from
a Raspberry Pi
Pico - tiny robots
ahoy!**

Solder this SHIM to the
back of your Pico and
unlock the magical ability

to **plug in up to two of our micro metal gearmotors** (using these tidy [short motor cables](#)). To connect them up via cable, your motors will need to be equipped with [Motor Connector Shims](#) (but if you'd rather not solder the shims to the motors yourself, we also sell them [with motors pre-attached](#)).

In addition to the motor driver, Motor SHIM for Pico features a multi-purpose and ever popular **user button** plus a **Qw/ST connector** so you can attach breakouts and give your diminutive robot some smarts. You could use it to make a tiny front wheel drive robot (like a mini [Trilobot](#)) or even a matchbox sized balancing robot.

We've designed it to be as slimline as possible and use minimal pins so it can be combined easily with other add-ons, like our [Pico Packs and Bases](#). For a smol battery powered robot with LiPo charging and an on/off button, consider using a [Pimoroni Pico LiPo](#) or stack this SHIM with a [LiPo SHIM for Pico](#)!

Motor SHIM for Pico is supported by a well documented C++/MicroPython motor library with lots of examples - there's even one to teach your motors how to sing! 🎵



Micro Metal Gearmotor Bracket (pack of 2) + £3.90



Micro Metal Gearmotor to LEGO® Axle Adaptor (pack of 4) + £3.90

Add to cart

You can adjust quantities once added to your cart.

Bulk or custom order? Go to our [wholesale store](#)! ❤️

Shop with confidence – we've been serving the hobbyist electronics, Maker, and retro gaming communities since 2012.

- Satisfaction or refund guarantee
- Worldwide shipping via mail or courier
- 57,000+ customer reviews
- Secure website and payments

"Female header and underneath your pico leaves just two cables for the motors."



JC in Assen, Netherlands

- Dual H-Bridge motor driver (DRV8833)
- 2 JST-ZH connectors (2 pin) for attaching motors
- User button
- Qw/ST (Qwiic/STEMMA QT) connector for attaching breakouts
- Compatible with Raspberry Pi Pico.
- Soldering required.
- [C++/MicroPython libraries](#)
- [Schematic](#)

Raspberry Pi Pico, headers and other components shown are sold separately.

Assembly

You'll need to solder the SHIM to the back of your Pico, with the mounting holes lining up at the USB port end. The 'PICO MOTOR SHIM' text on the SHIM and the pin labels on the back of the Pico should be facing each other.

Software

Our **C++/MicroPython** libraries are packed with powerful features for working with motors. You'll get best performance using C++, but if you're a beginner we'd recommend using our batteries included MicroPython

build for ease of getting started.

- [Download Pirate brand MicroPython](#)
- [MicroPython API documentation](#)
- [MicroPython examples](#)
- [C++ examples](#)

Alternatively, you can use **CircuitPython** to drive motors with your Raspberry Pi Pico, if you want access to all the nice conveniences of Adafruit's ecosystem.

- [Getting Started with CircuitPython](#)
- [CircuitPython example](#)

Notes

- Dimensions: approx 26mm x 21mm x 5mm (L x W x H, including connectors)
- If you're ambitious in the ways of experimental soldering, you can try soldering the Pico and the SHIM both to the short end of your header, back to back - leaving the long ends of the pins free to plug into other things. This method makes for a much more slimline Pico/SHIM package, but you'll need to make sure your solder joints make

good contact with the pads of both boards and the header.

- If you'd prefer not to attach Motor SHIM permanently to your Pico, consider soldering it to a pair of [socket headers](#) (as pictured in the photos of Pimoroni Pico LiPo).
- The little pink robot weighs just 68.8g! It uses Motor SHIM for Pico, [LiPo SHIM for Pico](#), a [1/2" ball caster](#), [white motor brackets](#), [32x7mm wheels](#) and a dinky [150mAh LiPo battery](#). The chassis baseplate is laser cut from 3mm acrylic ([download the .dxf file](#)).

About Raspberry Pi Pico

[Raspberry Pi Pico](#) is a flexible, low cost **microcontroller development board** from the folks at Raspberry Pi, based on their very own chip - the RP2040. It's easily programmable over USB with C/C++ or MicroPython, and ideal for using in all sorts of physical computing projects, devices and inventions - we're so excited to see what you make with it!

We've called our Pico-sized add-ons **packs**, as they're designed to attach to the back of your Pico as if it were wearing a very stylish back pack (or a miniature jet pack, if you prefer). We've also got Pico **bases** (larger add-on boards with a space to mount your Pico on top) and some other boards that let you do interesting hackerly things like using multiple packs at once - [click here](#) to view them all!

Your Pico will need to have male headers soldered to it (with the pins pointing downwards) to attach to our add-on boards.

7 customer reviews



9 months ago



by Guido about [Motor SHIM for Pico](#) via [REVIEWS.io](#)



3 years ago

Tiny and perfect for what I needed



by Daryl about [Motor SHIM for Pico](#) via [REVIEWS.io](#)



4 years ago

Impressed with how simple the Motor Shim was to get up and running. Clearer documentation on the button would have been useful, I had to refer to the schematic to figure

by Andrew about [Motor SHIM for Pico](#) via [REVIEWS.io](#)



4 years ago

Work really well and easy to use with Micropython.



by Sarrailh about [Motor SHIM for Pico](#) via [REVIEWS.io](#)



4 years ago

Products arrived fast, with good service from the people there.



by Dennis about [Motor SHIM for Pico](#) via [REVIEWS.io](#)



4 years ago

Female header and underneath your pico leaves just two cables for



by JC about [Motor SHIM for Pico](#) via [REVIEWS.io](#)



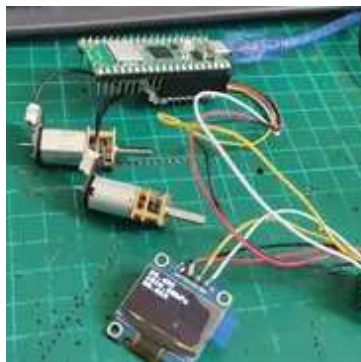
4 years ago

Easy to use, drives the motors consistantly and well made.



by Gary about [Motor SHIM for Pico](#) via [REVIEWS.io](#)

1 user photo



SUBSCRIBE FOR UPDATES

We'll send you information about new product launches and special offers!

THE GOOD SHIP PIMORONI

We've been sailing the Maker seas offering our wares since 2012.

- [read about our history](#)
- [what our customers say](#)
- [check out our products](#)

Follow us on social media:
[Bluesky](#),
[YouTube](#),
[Facebook](#),
[Instagram](#).

INFORMATION

Not sure about something?
Check out the links below.

- [shipping information](#)
- [ask a question in our forums](#)
- [contact our support team](#)
- [find a distributor](#)

FREQUENTLY ASKED QUESTIONS

Often it's the simplest of things.

- [how can I track my order?](#)
- [cost and time to ship to ...?](#)
- [are you VAT registered in the EU?](#)
- [can I collect my order?](#)

View other [frequently asked questions](#).

PIMORONI LTD.

Unit 1, Parkway Business Park, Parkway Drive, Sheffield, S9 4WN, United Kingdom

Registered in England and Wales Company №: 07510759 VAT Reg №: GB 140 8707 21

[Terms & Conditions](#) | [Privacy Policy](#) | [Cellular Service T&Cs](#)